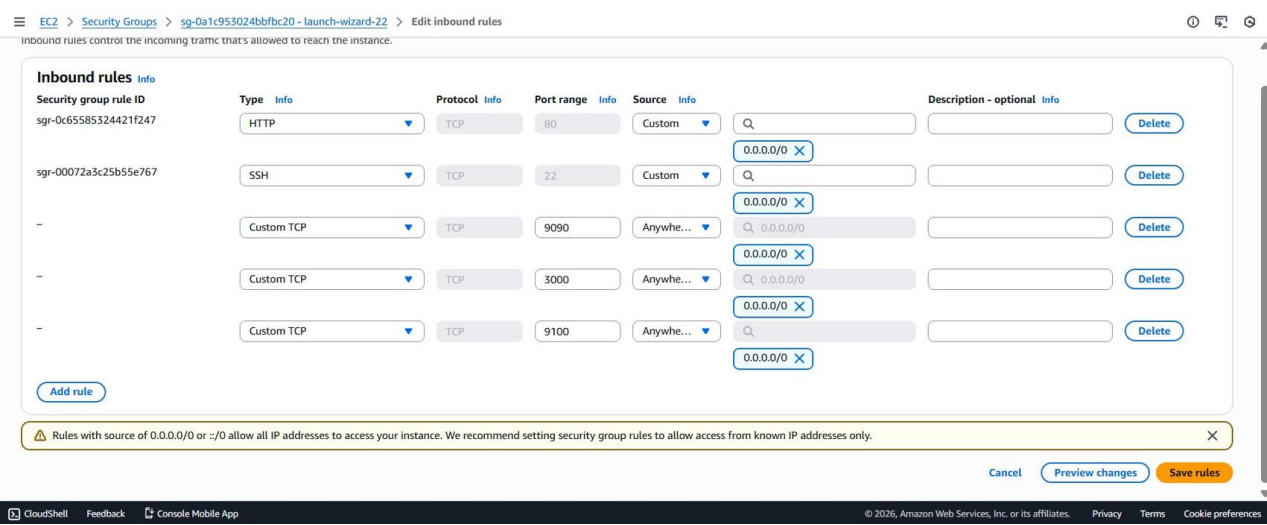


Task Description:

Install Prometheus and Grafana on a Linux EC2 machine, connect Prometheus to Grafana, and create a dashboard to view metrics.

Output:



```
ubuntu@ip-172-31-15-63: ~
Sourcing file '/etc/default/grub.d/50-cloudimg-settings.cfg'
Generating grub configuration file ...
GRUB_FORCE_PARTUUID is set, will attempt initrdless boot
Found linux image: /boot/vmlinuz-6.14.0-1018-aws
Found initrd image: /boot/microcode.cpio /boot/initrd.img-6.14.0-1018-aws
Found linux image: /boot/vmlinuz-6.14.0-1015-aws
Found initrd image: /boot/microcode.cpio /boot/initrd.img-6.14.0-1015-aws
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
6.14.0-1015-aws
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.14.0-1018-aws.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...
systemctl restart irqbalance.service multipathd.service polkit.service rsyslog.service ssh.service udisks2.service

Service restarts being deferred:
systemctl restart ModemManager.service
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
ubuntu @ session #3: apt[1563], sshd[1090]
ubuntu @ user manager service: systemd[1095]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-15-63:~$
```

```
ubuntu@ip-172-31-15-63: /opt
No containers need to be restarted.

User sessions running outdated binaries:
ubuntu @ session #3: apt[1563], sshd[1090]
ubuntu @ user manager service: systemd[1095]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-15-63:~$ cd /opt
ubuntu@ip-172-31-15-63:/opt$ sudo wget https://github.com/prometheus/node_exporter/releases/download/v1.7.0/node_exporter-1.7.0.linux-amd64.tar.gz
--2026-01-02 11:49:29-- https://github.com/prometheus/node_exporter/releases/download/v1.7.0/node_exporter-1.7.0.linux-amd64.tar.gz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)[20.207.73.82]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/9524057/01323270-6ecb-47aa-813f-52a2a89cdc64?sp=r&sv=2018-11-09&sr=b&
pr=https&se=2026-01-02T12%3A23%3A32Z&rsct=attachment%3B%3D%3A23%3A32Z&skv=2018-11-09&sig=K8f4W7z
43al-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4deskt=2026-01-02T11%3A22%3A56Z&skv=2018-11-09&sig=K8f4W7z
aXkuRf0EtIXze4w7BpGcwWSrHyv0wtIMJc3D6jwT=eyJ0eXAI0iJKVlQlLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiaXVkeiJoiOiJmVjZlZS1hc3N1dHMuZ210aHViZmVud21uZG93cy5uZXQ1fQ.AXII9C1xZW
bmQuY29tIiwia2V5IjoiZmVjZS1hc3N1dHMuZ210aHViZmVud21uZG93cy5uZXQ1fQ.AXII9C1xZW
hMG01H1KD5naFQxlpVgQ0cK6itemJDF_E&response-content-disposition=attachment%3B%20filename%3Dnode_exporter-1.7.0.linux-amd64.tar.gz&response-content-type=applic
ation%2Foctet-stream [following]
--2026-01-02 11:49:30-- https://release-assets.githubusercontent.com/github-production-release-asset/9524057/01323270-6ecb-47aa-813f-52a2a89cdc64?sp=r&sv=20
18-11-09&sr=b&pr=https&se=2026-01-02T12%3A23%3A32Z&rsct=attachment%3B%3D%3A23%3A32Z&skv=2018-11-09&sig=K8f4W7z
43al-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4deskt=2026-01-02T11%3A22%3A56Z&skv=2018-11-09&sig=K8f4W7z
aXkuRf0EtIXze4w7BpGcwWSrHyv0wtIMJc3D6jwT=eyJ0eXAI0iJKVlQlLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiaXVkeiJoiOiJmVjZlZS1hc3N1dHMuZ210aHViZmVud21uZG93cy5uZXQ1fQ.AXII9C1xZW
hMG01H1KD5naFQxlpVgQ0cK6itemJDF_E&response-content-disposition=attachment%3B%20filename%3Dnode_exporter-1.7.0.linux-amd64.tar.gz&response-cont
ent-type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)[185.199.108.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 10419253 (9.9M) [application/octet-stream]
Saving to: 'node_exporter-1.7.0.linux-amd64.tar.gz'

node_exporter-1.7.0.linux-amd64.tar.gz 100%[=====>] 9.94M --.-KB/s in 0.02s

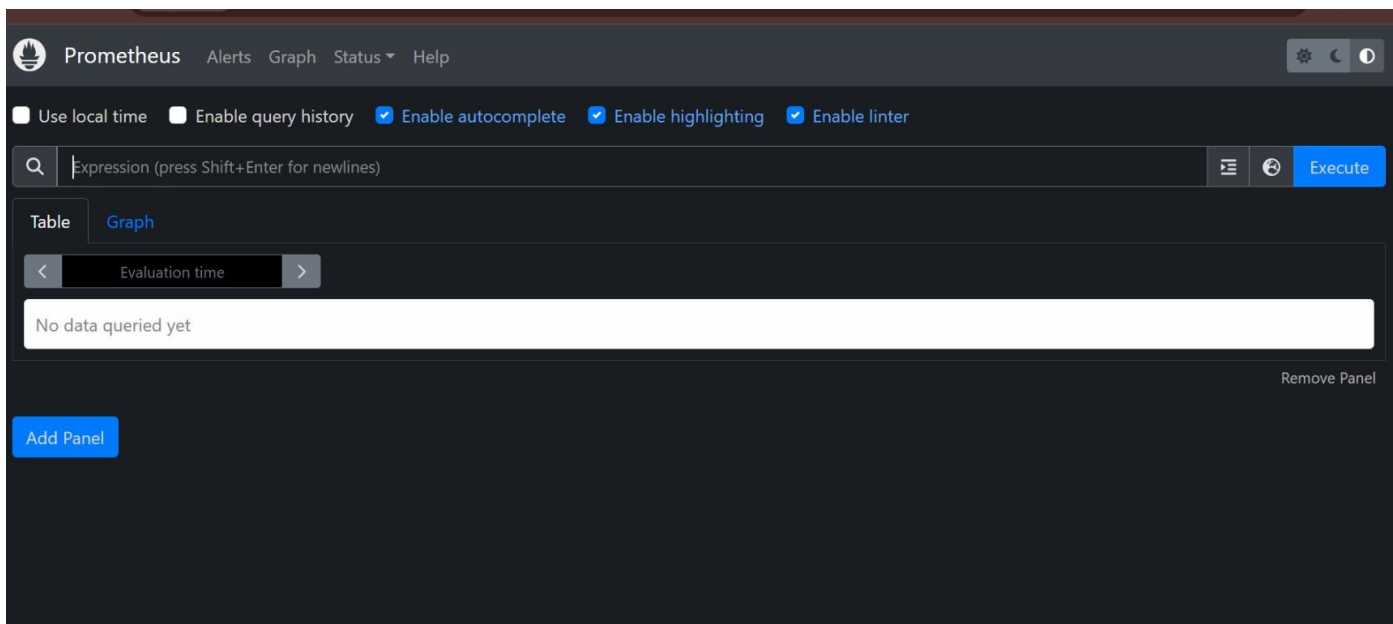
2026-01-02 11:49:30 (466 MB/s) - 'node_exporter-1.7.0.linux-amd64.tar.gz' saved [10419253/10419253]

ubuntu@ip-172-31-15-63:/opt$ sudo tar -xvf node_exporter-1.7.0.linux-amd64.tar.gz
node_exporter-1.7.0.linux-amd64/
node_exporter-1.7.0.linux-amd64/LICENSE
node_exporter-1.7.0.linux-amd64/node_exporter
node_exporter-1.7.0.linux-amd64/NOTICE
ubuntu@ip-172-31-15-63:/opt$ sudo mv node_exporter-1.7.0.linux-amd64 node_exporter
ubuntu@ip-172-31-15-63:/opt$
```

```
PuTTY (inactive)
ubuntu@ip-172-31-15-63:/opt$ cd /opt/prometheus
sudo nano prometheus.yml
ubuntu@ip-172-31-15-63:/opt/prometheus$ ./prometheus --config.file=prometheus.yml
ts=2026-01-02T11:58:19.737Z caller=main.go:564 level=info msg="No time or size retention was set so using the default time retention" duration=15d
ts=2026-01-02T11:58:19.737Z caller=main.go:608 level=info msg="Starting Prometheus Server" mode=server version="(version=2.50.0, branch=HEAD, revision=814b92
0e8a6345d35712b5857ebd4cb5e90fc107)"
ts=2026-01-02T11:58:19.738Z caller=main.go:613 level=info build_context="(go=go1.21.7, platform=linux/amd64, user=root@384077elcf50, date=20240222-09:38:19,
tags=netgo,builtinassets,stringlabels)"
ts=2026-01-02T11:58:19.738Z caller=main.go:614 level=info host_details="(Linux 6.14.0-1015-aws #15-24.04.1-Ubuntu SMP Tue Sep 23 22:44:48 UTC 2025 x86_64 ip-
172-31-15-63 (none))"
ts=2026-01-02T11:58:19.738Z caller=main.go:615 level=info fd_limits="(soft=1048576, hard=1048576)"
ts=2026-01-02T11:58:19.738Z caller=main.go:616 level=info vm_limits="(soft=unlimited, hard=unlimited)"
ts=2026-01-02T11:58:19.740Z caller=query_logger.go:115 level=error component=activeQueryTracker msg="Failed to create directory for logging active queries"
ts=2026-01-02T11:58:19.740Z caller=query_logger.go:93 level=error component=activeQueryTracker msg="Error opening query log file" file=/opt/prometheus/data/q
ueries.active err="open data/queries.active: no such file or directory"
panic: Unable to create mmap-ed active query log

goroutine 1 [running]:
github.com/prometheus/prometheus/promql.NewActiveQueryTracker({0x3711e9d, 0x5}, 0x14, {0x4089800, 0xc000735a40})
/app/promql/query_logger.go:123 +0x411
main.main()
/app/cmd/prometheus/main.go:739 +0x8252
ubuntu@ip-172-31-15-63:/opt/prometheus$ sudo nano prometheus.yml
ubuntu@ip-172-31-15-63:/opt/prometheus$ ./prometheus --config.file=prometheus.yml
ts=2026-01-02T12:00:49.737Z caller=main.go:564 level=info msg="No time or size retention was set so using the default time retention" duration=15d
ts=2026-01-02T12:00:49.737Z caller=main.go:608 level=info msg="Starting Prometheus Server" mode=server version="(version=2.50.0, branch=HEAD, revision=814b92
0e8a6345d35712b5857ebd4cb5e90fc107)"
ts=2026-01-02T12:00:49.737Z caller=main.go:613 level=info build_context="(go=go1.21.7, platform=linux/amd64, user=root@384077elcf50, date=20240222-09:38:19,
tags=netgo,builtinassets,stringlabels)"
ts=2026-01-02T12:00:49.737Z caller=main.go:614 level=info host_details="(Linux 6.14.0-1015-aws #15-24.04.1-Ubuntu SMP Tue Sep 23 22:44:48 UTC 2025 x86_64 ip-
172-31-15-63 (none))"
ts=2026-01-02T12:00:49.737Z caller=main.go:615 level=info fd_limits="(soft=1048576, hard=1048576)"
ts=2026-01-02T12:00:49.737Z caller=main.go:616 level=info vm_limits="(soft=unlimited, hard=unlimited)"
ts=2026-01-02T12:00:49.739Z caller=query_logger.go:115 level=error component=activeQueryTracker msg="Failed to create directory for logging active queries"
ts=2026-01-02T12:00:49.739Z caller=query_logger.go:93 level=error component=activeQueryTracker msg="Error opening query log file" file=/opt/prometheus/data/q
ueries.active err="open data/queries.active: no such file or directory"
panic: Unable to create mmap-ed active query log

goroutine 1 [running]:
github.com/prometheus/prometheus/promql.NewActiveQueryTracker({0x3711e9d, 0x5}, 0x14, {0x4089800, 0xc0009e6e10})
/app/promql/query_logger.go:123 +0x411
main.main()
/app/cmd/prometheus/main.go:739 +0x8252
ubuntu@ip-172-31-15-63:/opt/prometheus$
```



```
ubuntu@ip-172-31-15-63: /opt/node_exporter
Selecting previously unselected package grafana.
Preparing to unpack ../grafana_12.3.1_amd64.deb ...
Unpacking grafana (12.3.1) ...
Setting up musl:amd64 (1.2.4-2) ...
Setting up grafana (12.3.1) ...
info: Selecting UID from range 100 to 999 ...

info: Adding system user 'grafana' (UID 111) ...
info: Adding new user 'grafana' (UID 111) with group 'grafana' ...
info: Not creating home directory '/usr/share/grafana'.
### NOT starting on installation, please execute the following statements to configure grafana to start automatically using systemd
sudo /bin/systemctl daemon-reload
sudo /bin/systemctl enable grafana-server
### You can start grafana-server by executing
sudo /bin/systemctl start grafana-server
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
 6.14.0-1015-aws
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.14.0-1018-aws.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-15-63:/opt/node_exporter$
```

```

ubuntu@ip-172-31-15-63: ~
ubuntu@ip-172-31-15-63:~$ history
1 clear
2 sudo apt update -y
3 sudo apt upgrade -y
4 cd /opt
5 sudo wget https://github.com/prometheus/node_exporter/releases/download/v
1.7.0/node_exporter-1.7.0.linux-amd64.tar.gz
6 sudo tar -xvf node_exporter-1.7.0.linux-amd64.tar.gz
7 sudo mv node_exporter-1.7.0.linux-amd64 node_exporter
8 cd /opt/node_exporter
9 ./node_exporter
10 clear
11 ps -ef | grep prometheus
12 sudo netstat -tulnp | grep 9090
13 cd /opt/prometheus
14 ls
15 sudo netstat -tulnp | grep 9090
16 /opt/prometheus
17 ./prometheus --config.file=prometheus.yml
18 sudo apt install net-tools -y
19 ./prometheus --config.file=prometheus.yml
20 mkdir data
21 sudo mkdir data
22 mkdir data
23 sudo mkdir /opt/prometheus/data
24 sudo chmod 755 /opt/prometheus/data
25 ls -ld /opt/prometheus/data
26 cd /opt/prometheus
27 ./prometheus --config.file=prometheus.yml
28 sudo mkdir -p /opt/prometheus/data
29 sudo chown -R ubuntu:ubuntu /opt/prometheus
30 cd /opt/prometheus
31 ./prometheus --config.file=prometheus.yml --storage.tsdb.path=/opt/pr
ometheus/data --web.listen-address="0.0.0.0:9090"
32 clear
33 history
ubuntu@ip-172-31-15-63:~$

```

```

ubuntu@ip-172-31-15-63: /opt/node_exporter

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

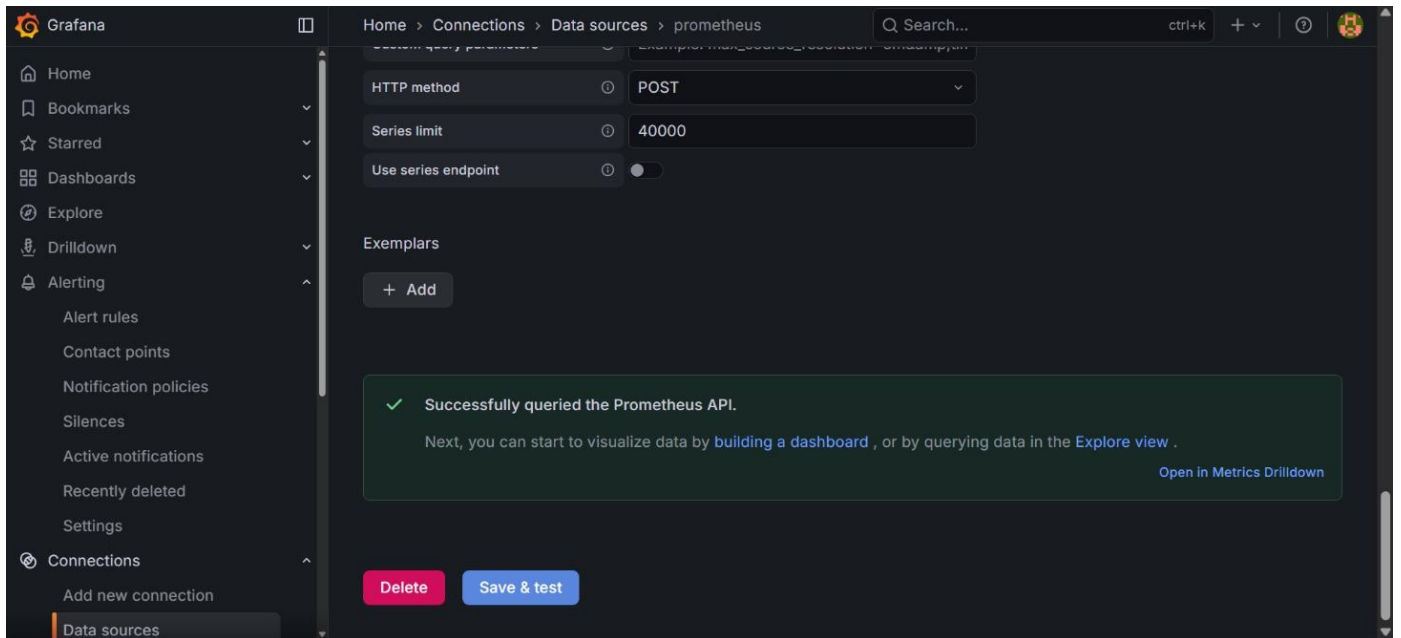
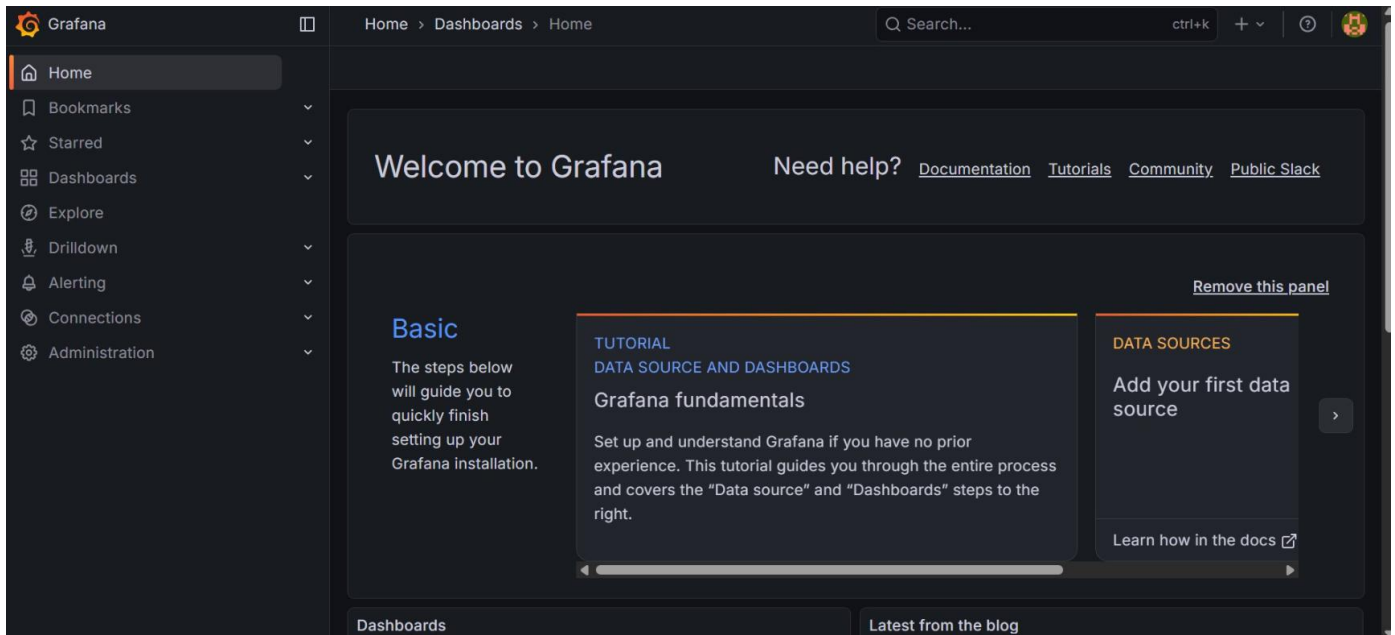
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-15-63:/opt/node_exporter$ sudo systemctl start grafana-server
sudo systemctl enable grafana-server
Synchronizing state of grafana-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable grafana-server
Created symlink /etc/systemd/system/multi-user.target.wants/grafana-server.service → /usr/lib/systemd/system/grafana-server.service.
ubuntu@ip-172-31-15-63:/opt/node_exporter$ sudo systemctl status grafana-server
● grafana-server.service - Grafana instance
   Loaded: loaded (/usr/lib/systemd/system/grafana-server.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-01-02 12:24:08 UTC; 13s ago
     Docs: http://docs.grafana.org
    Main PID: 14957 (grafana)
      Tasks: 16 (limit: 1008)
    Memory: 331.8M (peak: 347.2M)
       CPU: 4.225s
    CGroup: /system.slice/grafana-server.service
            └─14957 /usr/share/grafana/bin/grafana server --config=/etc/grafana/grafana.ini --pidfile=/run/grafana/grafana-server.pid --packaging=deb cfg:

Jan 02 12:24:19 ip-172-31-15-63 grafana[14957]: logger=plugin.backgroundinstaller t=2026-01-02T12:24:19.297373369Z level=info msg="Installing plugin" plugin
Jan 02 12:24:19 ip-172-31-15-63 grafana[14957]: logger=plugin.installer t=2026-01-02T12:24:19.893893972Z level=info msg="Installing plugin" pluginId=grafana>
Jan 02 12:24:19 ip-172-31-15-63 grafana[14957]: logger=installer.fs t=2026-01-02T12:24:19.967125696Z level=info msg="Downloaded and extracted grafana-explor>
Jan 02 12:24:20 ip-172-31-15-63 grafana[14957]: logger=plugins.registration t=2026-01-02T12:24:20.007428141Z level=info msg="Plugin registered" pluginId=gra>
Jan 02 12:24:20 ip-172-31-15-63 grafana[14957]: logger=plugin.backgroundinstaller t=2026-01-02T12:24:20.007488295Z level=info msg="Plugin successfully insta>
Jan 02 12:24:20 ip-172-31-15-63 grafana[14957]: logger=plugin.backgroundinstaller t=2026-01-02T12:24:20.007522126Z level=info msg="Installing plugin" plugin>
Jan 02 12:24:20 ip-172-31-15-63 grafana[14957]: logger=plugin.installer t=2026-01-02T12:24:20.758631372Z level=info msg="Installing plugin" pluginId=grafana>
Jan 02 12:24:20 ip-172-31-15-63 grafana[14957]: logger=installer.fs t=2026-01-02T12:24:20.840496132Z level=info msg="Downloaded and extracted grafana-metric>
Jan 02 12:24:20 ip-172-31-15-63 grafana[14957]: logger=plugins.registration t=2026-01-02T12:24:20.883373124Z level=info msg="Plugin registered" pluginId=gra>
Jan 02 12:24:20 ip-172-31-15-63 grafana[14957]: logger=plugin.backgroundinstaller t=2026-01-02T12:24:20.883482919Z level=info msg="Plugin successfully insta>
ubuntu@ip-172-31-15-63:/opt/node_exporter$

```

```
Putty (inactive)
Jan 02 12:24:20 ip-172-31-15-63 grafana[14957]: logger-plugins.registration t=2026-01-02T12:24:20.883373124Z level=info msg="Plugin registered" pluginId=grafana
Jan 02 12:24:20 ip-172-31-15-63 grafana[14957]: logger-plugin.backgroundinstaller t=2026-01-02T12:24:20.883482919Z level=info msg="Plugin successfully installed"
ubuntu@ip-172-31-15-63:/opt/node_exporter$ sudo grafana-cli admin reset-admin-password admin123
INFO [01-02|12:28:43] Starting Grafana logger=settings version=12.3.1 commit=0d1a5b4420a5e4579b91c239ecb240ea2b247fba branch=release-12.3.1 compiled=2026-01-02T12:28:43Z
INFO [01-02|12:28:43] Config loaded from logger=settings file=/usr/share/grafana/conf/defaults.ini
INFO [01-02|12:28:43] Config loaded from logger=settings file=/etc/grafana/grafana.ini
INFO [01-02|12:28:43] Config overridden from command line logger=settings arg="default.paths.data=/var/lib/grafana"
INFO [01-02|12:28:43] Config overridden from command line logger=settings arg="default.paths.logs=/var/log/grafana"
INFO [01-02|12:28:43] Config overridden from command line logger=settings arg="default.paths.plugins=/var/lib/grafana/plugins"
INFO [01-02|12:28:43] Config overridden from command line logger=settings arg="default.paths.provisioning=/etc/grafana/provisioning"
INFO [01-02|12:28:43] Target logger=settings target=[all]
INFO [01-02|12:28:43] Path Home logger=settings path=/usr/share/grafana
INFO [01-02|12:28:43] Path Data logger=settings path=/var/lib/grafana
INFO [01-02|12:28:43] Path Logs logger=settings path=/var/log/grafana
INFO [01-02|12:28:43] Path Plugins logger=settings path=/var/lib/grafana/plugins
INFO [01-02|12:28:43] Path Provisioning logger=settings path=/etc/grafana/provisioning
INFO [01-02|12:28:43] App mode production logger=settings
INFO [01-02|12:28:43] FeatureToggles logger=featuremgmt dashboardDsAdHocFiltering=true ssoSettingsLDAP=true alertingBulkActionsInUI=true pinNavItems=true onPremToCloudMigrations=true prometheusAzureOverrideAudience=true logsContextDatasourcesUi=true logsExploreTableVisualisation=true unifiedStorageHistoryPruner=true addFieldFromCalculationStatFunctions=true dashboardScene=true newFiltersUI=true preinstallAutoUpdate=true sharingDashboardImage=true awsAsyncQueryCaching=true cloudWatchCrossAccountQuerying=true azureMonitorEnableUserAuth=true improvedExternalSessionHandlingSAML=true alertRuleRestore=true annotationPermissionUpdate=true newPDFRendering=true alertingUseNewSimplifiedRoutingHashAlgorithm=true kubernetesDashboards=true alertingRuleRecoverDeleted=true lokiQuerySplitting=true dataplaneFrontendFallback=true alertingMigrationUI=true transformationsRedesign=true tlsMemcached=true cloudWatchRoundUpEndTime=true cloudWatchNewLabelParsing=true correlations=true logsInfiniteScrolling=true alertingUIOptimizeReducer=true azureMonitorPrometheusExemplars=true useSessionStorageForRedirection=true influxdbBackendMigration=true grafanaAssistantInProfilesDrilldown=true publicDashboardsScene=true grafanaconThemes=true alertingNotificationsStepMode=true dashboardSceneSolo=true alertingQueryAndExpressionsStepMode=true awsDatasourcesTempCredentials=true alertingSaveStateCompression=true dashboardSceneForViewers=true improvedExternalSessionHandling=true unifiedRequestLog=true logsPanelControls=true newLogsPanel=true panelMonitoring=true dashboardpt=true preventPanelChromeOverflow=true alertingRuleVersionHistoryRestore=true alertingImportYAMLUI=true onlyStoreActionSets=true recordedQueriesMulti=true formatString=true logRowsPopoverMenu=true alertingRulePermanentlyDelete=true adhocFiltersInTooltips=true lokiLabelNamesQueryApi=true groupToNestedTableTransformation=true
INFO [01-02|12:28:43] Connecting to DB logger=sqlstore dbtype=sqlite3
INFO [01-02|12:28:43] Using SQLite driver logger=sqlstore driver="mattn/go-sqlite3 (CGO enabled)"
INFO [01-02|12:28:43] Locking database logger=migrator
INFO [01-02|12:28:43] Starting DB migrations logger=migrator
INFO [01-02|12:28:43] migrations completed logger=migrator performed=0 skipped=689 duration=1.060238ms
INFO [01-02|12:28:43] Unlocking database logger=migrator
INFO [01-02|12:28:43] Envelope encryption state logger=secrets enabled=true current provider=secretKey.v1

Admin password changed successfully ✓

ubuntu@ip-172-31-15-63:/opt/node_exporter$
```